

CULPEPER, VA, USA

WMO: 723098

Lat: 38.527N

Lon: 77.859W

Elev: 96

StdP: 100.17

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 93798

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.1	-7.9	-20.1	0.6	-5.3	-17.2	0.8	-4.1	8.8	4.6	8.0	4.2	0.8	0	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	34.8	24.7	33.1	24.1	32.3	24.0	27.0	31.3	26.1	30.7	25.3	30.2	2.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.7	29.1	24.8	20.1	28.2	23.9	19.0	27.3	85.5	31.4	81.2	30.4	77.6	30.2	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.5	5.4	DB	-15.4	36.6	2.6	1.8	-17.3	37.9	-18.8	39.0	-20.3	40.0	-22.1	41.4	(4)
(5)			WB	-15.7	28.1	2.6	1.2	-17.6	28.9	-19.1	29.6	-20.6	30.3	-22.4	31.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	1.8	3.5	8.0	13.9	18.8	23.1	25.3	24.6	20.9	14.5	8.6	3.8
	DBStd	9.37	5.75	5.32	5.26	4.83	4.10	3.04	2.55	2.79	3.52	4.66	4.70	4.75
	HDD10.0	864	262	192	103	16	0	0	0	0	0	11	81	200
	HDD18.3	2344	512	416	321	149	46	3	0	0	16	138	294	449
	CDD10.0	2311	8	9	42	132	273	393	474	454	327	150	38	9
	CDD18.3	748	0	0	1	16	61	146	216	196	93	18	1	0
	CDH23.3	7439	1	4	34	254	648	1441	2178	1854	813	195	16	1
CDH26.7	3071	0	0	4	80	224	589	1008	817	288	58	2	0	
Wind	WSAvg	1.6	1.8	2.0	2.2	2.1	1.5	1.3	1.1	1.1	1.2	1.4	1.5	1.5
Precipitation	PrecAvg	1048	75	55	97	83	103	101	103	83	104	80	81	81
	PrecMax	1433	168	166	229	126	208	213	180	185	259	205	157	146
	PrecMin	746	25	15	0	40	35	35	34	31	0	2	22	40
	PrecStd	157	40	31	50	27	47	45	41	36	66	51	41	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.0	22.9	27.2	32.3	32.9	35.9	37.3	36.4	34.1	31.4	25.8	21.3
		MCWB	16.2	14.8	16.9	19.7	22.5	24.9	25.8	24.7	23.6	22.5	18.0	15.4
	2%	DB	17.7	18.8	23.6	28.9	31.3	33.0	35.0	34.1	32.3	27.8	22.4	17.6
		MCWB	13.8	12.7	14.9	18.4	22.2	24.1	25.1	24.7	23.2	20.0	16.2	14.1
	5%	DB	14.1	16.0	21.0	26.6	29.0	32.1	33.2	32.7	29.8	25.2	19.2	14.3
		MCWB	9.7	10.0	13.8	17.2	21.3	24.0	24.7	24.4	22.4	19.0	14.8	10.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	10%	DB	11.1	12.5	17.8	23.4	27.4	30.3	32.2	31.2	27.8	22.7	17.4	12.2
		MCWB	7.6	7.8	11.9	16.0	20.4	23.0	24.5	23.8	21.3	17.7	13.0	8.9
	0.4%	WB	17.7	16.8	18.7	21.8	25.2	27.4	28.0	28.0	26.6	23.7	20.4	18.3
		MCDB	19.6	21.2	22.9	26.3	30.9	30.9	32.7	32.2	29.7	27.8	22.6	19.3
	2%	WB	14.8	13.8	16.8	20.1	23.8	26.1	27.1	26.8	25.0	22.3	18.0	15.1
		MCDB	17.5	17.7	21.5	26.2	28.5	31.0	31.7	31.1	29.2	26.2	20.5	17.7
Mean Daily Temperature Range	5% DB	WB	10.9	10.9	14.9	18.7	22.7	25.1	26.1	25.8	23.9	20.9	16.0	11.9
		MCDB	13.5	14.3	19.7	24.6	27.5	30.4	31.0	30.2	27.2	23.5	18.7	13.6
	5% WB	WB	7.9	8.3	12.8	17.2	21.5	24.1	25.3	25.0	22.9	18.9	13.7	9.0
		MCDB	10.3	12.0	17.1	21.7	25.6	28.6	30.4	29.2	26.2	21.8	16.4	11.5
Clear-Sky Solar Irradiance	taub	0.300	0.317	0.353	0.394	0.434	0.490	0.507	0.509	0.418	0.356	0.320	0.305	
	taud	2.499	2.432	2.390	2.299	2.211	2.094	2.072	2.044	2.281	2.452	2.510	2.520	
	Ebn at Noon	884	910	907	884	851	801	783	769	827	854	852	851	
	Edh at Noon	78	95	110	128	142	160	162	163	121	93	77	72	
All-Sky Solar Radiation	RadAvg	2.14	2.96	3.94	4.92	5.51	6.07	5.88	5.28	4.35	3.32	2.43	1.89	
	RadStd	0.19	0.33	0.25	0.37	0.54	0.43	0.32	0.26	0.52	0.37	0.20	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (60 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+128	+76

Nomenclature: See separate page